

**CEC 314: TRANSPORTATION ENGINEERING 3 UNITS****DEFINATION**

Transportation Engineering is the application of technology and scientific principles to the planning, functional design, operation and management facilities for any mode of transportation in order to provide for the safe, rapid, comfortable, convenient, economical and environmentally compactable movement of people and goods.

**IMPORTANCE OF TRANSPORTATION****1. ROLE OF TRANSPORTATION**

Transportation contributes to the economic, industrial, social and cultural development of any country. Transportation is vital for the economic development of any region because whatever every community produces whether food, clothing, industrial products or medicine needs transport at all stages from the production to distribution. In the production stage, transportation is required for conveying raw materials like seeds, manure, coal, steel, etc to where they will be processed. In the distribution stage, transportation is required from the production centres e.g. farms and factories to the marketing centres and later to the retailers and the consumers for distribution. Inadequate transportation facilities retard the process of socio-economic development of any country.

**2. ECONOMIC ACTIVITY AND TRANSPORTATION**

The economic activities are the processes by means of which the products are utilized by man to satisfy his wants. The important factors of economic activity are demand and supply. The effect of transport on economic activity is that if the transportation system is efficient, then the cost of products can be lower. The transportation cost is always an influencing factor on consumer price of commodities.

**3. SOCIAL EFFECTS OF TRANSPORTATION.**

Population has always settled along river shores, roadsides and near railway stations. In the present concept of transportation network, this kind of ribbon development is greatly discouraged. Attempts are being made to decentralize the population centres away in such a way that they are avoiding overcrowding around the populated areas by developing suburban areas.

**4. ROLE OF TRANSPORTATION IN RURAL DEVELOPMENT**

With over 70% of the population of the country living in the rural villages, the development of urban centers alone does not indicate the overall development of the country. Only with the development in transportation facilities in rural areas, there could be faster development of the rural areas. Things like fertilizers and other inputs for agriculture and small industries could

reach the rural population easily and similarly, the products can be sold at the nearest marketing centres for remunerative price resulting in faster economic growth and decreased wastage. With improved facilities for education, health care and other social needs in the villages, the urge for the migration to urban centres decreases, thus helping in balance development of the country as a whole.

### **ADVANTAGES OF TRANSPORTATION**

1. Transportation is for the development of the community
2. Transportation is essential for the economic prosperity and general development of the country.
3. Transportation is essential for strategic movement in emergency for defense of the country and to maintain better law and order.

### **MODES OF TRANSPORTATION**

There are three basic modes of transportation: land, water and air. Land has given scope for the development of road and rail transport. Water and air have developed water ways and air ways respectively. The roads or highways not only include the modern highway system but also the city streets, feeder roads and village roads, catering for a wide range of road vehicles and the pedestrians. Railway has been developed for long distance transportation and for urban travel. Water ways include oceans, rivers, canals and lakes for the movement of ships and boats.

The four major modes of transportation are

- Roadways or highways
- Railways
- Waterways
- Airways

Transportation by air is the fastest among the four groups. Air travels also provide more comfort apart from saving in transportation time for passengers and the goods between the airports. Transportation by water is the slowest mode of transportation but this mode needs minimum energy to haul a unit load through a unit distance. Transportation by water is only possible between ports on the sea routes or along rivers or canals where inland transportation facilities are available. The transportation along the rail track could be advantageous by railways between the stations both for passengers and goods, particularly for longer distances. These railway tracks could serve as arteries for transportation by land and the roads could serve as feeder system for transportation to the interior parts and to the intermediate localities between the railway stations. The cost of transportation required by rail is by far smaller ( $1/4$ ) of the cost of transportation required by road. Therefore, full advantage of this mode should be taken for the transportation of bulk goods along land where the transportation facilities are available.

Transportation by road is the only mode which could give maximum service to one or all. It also has maximum flexibility for travel with reference to route, direction, time and speed of travel, etc. through any mode of road vehicle. It is possible to provide door to door services only

by road transport. The other three (airways, waterways and railways) have to depend on road transportation for their services to and from their terminals, airports, harbours or stations.

### **CHARACTERISTICS OF ROAD TRANSPORT**

Road transport is the nearest mode of transportation to the people. The passengers and the goods have to be first transported by road before reaching a railway station, seaport or an airport. The road network alone could serve the remotest villages of a vast country like ours. The characteristics of road transport include:

- ✓ Roads are used by various types of vehicles like passenger cars, buses, trucks, two and three wheels automobiles, pedal cycles and animal drawn vehicles. But railways tracks are only used by rail locomotives and wagons, waterways are used by only ships and boats while airways are used only by planes and helicopters.
- ✓ Road transport requires only a small investment for the government. Motor vehicles are much cheaper than any other carrier like rail locomotives and wagons, water and air carriers. Construction and maintenance of roads are also cheaper than that of railway tracks, docks, harbours and airports.
- ✓ Road transportation offers a complete freedom to road users to transfer their vehicles from one lane to another and from one road to another according to the need and convenience. This flexibility of changes in location, direction, speed and timing of travel is not available to other modes of transportation.
- ✓ In particular for short distance travel, road transport saves time. Trains stop at junctions and main stations for comparatively longer time.
- ✓ Speed of movement is directly related to the severity of accidents. The road safety decreases with increasing dispersion of speed. Road transport is subjected to a high degree of accidents due to the flexibility of movements offered to the road users. Derailment of railway locomotives and air crash of air planes are also not common. They are in fact more dangerous if they occur.
- ✓ Road transport is the only means of transportation that offers itself to the whole community alike.

### **VOLUME STUDIES**

The number of vehicles passing a given section of road per given period of time is one of the fundamental measures of traffic on road system. It is termed as volume or flow and is expressed in vehicles per hour or vehicles per day. When the traffic is composed by a number of types of vehicles, it is more convenient to convert the flow into equivalent passenger car units (pcu) by using equivalent factors.

### TYPES OF COUNTS

1. **Annual Average Daily Traffic (AADT):** This is the yearly volume divided by the number of days in the year.
2. **Average Daily Traffic (ADT):** The total volume during a given time period in whole days and less than one year divided by the number of days in that time period.
3. **Maximum Annual hourly volume (MAHN):** The highest hourly volume that occurs on a roadway in a designated year.

### HIGHWAY TRAFFIC STREAM CHARACTERISTICS

The flow of vehicles along a highway is characterized by three fundamental traffic parameters: speed, volume and density.

1. **Volume speed:** Speed is the rate of travel expressed in Km/h. vehicle speed varies both in time and space and can be measured at a point or can be averaged over a relatively long section of street of highway between an origin and destination.

### TYPES OF VEHICLES SPEED

1. Spot speed: This is speed measured at a point or instantaneous speed.
2. Running speed: This is the distance travelled divided by the running time.
3. Journey speed: This is the total distance travelled divided by the total time of travel.

Methods of measuring spot speed

1. Radar speed meter and optical instrument
2. Electrical timing photography, video and records.

Methods of measuring traffic stream speed

1. Moving car observer method
2. Registration plate number method.
3. Elevated observer method

### EXAMPLE

Mr John left Mubi for Yola which are 178 km apart at 6:00am and by 7:15am he was at Hong. He stopped at Hong for 45 min for his breakfast and by 8:45, he arrived Gombi where he spent 30 min in order to refill his tank with fuel. It took him another 30 mins to reach Song where he spent 45mins with a friend gisting. From Song he finally reached his destination after 45 mins.

1. What was his running speed?
2. What was his journey speed?

### SOLUTION

First of all separate the running time from the journey time i.e.

Running time = time from Mubi to Hong = 6:00 to 7:15 = 75mins

Time from Hong to Gombi = 45mins

Time from Gombi to Song = 30mins

Time from Song to Yola = 45mins

Therefore total running time is = 75 + 45 + 30 + 45 = 195mins = 3.25hrs

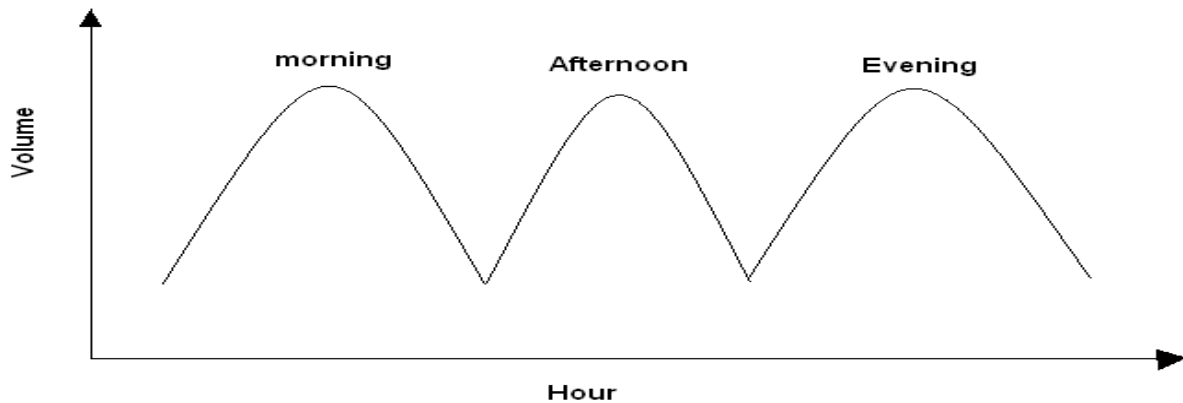
Journey time is the total time spent from Mubi to Yola i.e. = 75 + 45 + 45 + 30 + 30 + 45 + 45 = 315mins = 5.25hrs

$$\text{Speed} = \frac{\text{Distance}}{\text{Time}}$$

$$(a) \text{ Running speed} = \frac{\text{Distance}}{\text{Running Time}} = \frac{178\text{Km}}{3.25\text{hrs}} = 54.77\text{Km/hr}$$

$$(b) \text{ Journey speed} = \frac{\text{Distance}}{\text{Journey Time}} = \frac{178\text{Km}}{5.25\text{hr}} = 33.90\text{Km/hr}$$

**2. TRAFFIC VOLUME:** Traffic volume is defined as the number of vehicles that pass along a road way or traffic length per unit time. Traffic volume as a variable has a great importance to the traffic engineer. On a given roadway the volume of the traffic fluctuates widely. The variation that occurs tends to be cyclical and to some extend periodical.



A term rate of flow (or traffic flow) account for the variability of peaks that may occur during periods of that are less than one hour. The rate of flow is of great importance to highway analysis. The highway capacity manual uses the peak factor to relate peak rate of flow to hourly volume. The peak hour factor is defined as the ratio of total hourly volume to the maximum rate of flow within the hour. If there is no variation during the hour, the peak hour factor will be one (PHF=1)

$$\text{PHF} = \frac{\text{HOURLY VOLUME}}{\text{MAX. PEAK VOLUME IN THAT HOUR}}$$

Measures of traffic volume

- First, average daily traffic which is ADT

- Second, design hour volume (DHV). It is the feature of hourly volume that is used for design and is defined as the thirtieth highest hourly for volume for a design year.

Uses of volume data

1. It is used to establish the relative importance of any road.
2. To establish the influence of load on the road.
3. Used for highway classification and budget programming.
4. Used in traffic design mainly capacity required, accident control and traffic control.
5. Used in geometric design
6. Used in structural design of pavement.

## METHODS OF MEASUREMENT

1. **Manual method:** Use field personnel to count and classify traffic flowing pass a fixed point. The number of observers needed to count the vehicles depends upon the number of lanes in the highway on which the count is to be taken and the type of information desired. A desirable practice is to record traffic in each direction of travel separate and post observers for each direction. Mainly used when data cannot be obtained by mechanical means e.g. turning at intersections, occupancy, pedestrian, classification, and freeway count.
2. **Mechanical method:** Automatic device enable the count of traffic to be taken at any given location and a record to be kept of the count. They essentially consist of equipment for detecting the passage or presence of a vehicle. They always count number of axles and can be of permanent type and or a portable type. Permanent counters count on permanent basis while portable counters are for temporary or short term count e.g. pneumatic road tube.
3. **Combination of manual and mechanical method:** This involves the services of field personnel to operate the mechanical counter.
4. **Photographic method:** The method can be used for comprehensive traffic study, because ordinary manual count cannot fully cope up with the demand of a comprehensive study embracing different parameters such as volume, concentration, delay, speed, etc.
5. **Moving car observer method:** This method has been developed to provide simultaneous measurement of stream variables. It involves an observer who is taking certain measurements while moving in relation to the stream being measured. If  $N_o$  is the number of vehicles overtaking the driver and  $T$  is the period of observation, the observed flow is given by  $q = N_o/T$  when moving with the traffic stream.

$$\text{i.e. } q = \frac{N_o}{T} \dots\dots\dots(1)$$

If  $N_p$  is the number of vehicles overtaken by the driver and  $L$  is the distance travelled, density of flow  $K = \frac{N_p}{L} \dots\dots\dots(2)$

where  $L = VT$ ,  $V$  is the observer speed.

From eq. (1),  $N_o = qT$  and

From eq. (2),  $N_p = KVT$

The relative count,  $N = N_o - N_p$

$$= qT - KVT$$

$$\frac{N}{T} = q - KV \dots\dots\dots(3)$$

Eq. (3) is the basic equation for the moving car observer method which relates the stream variables  $V$  and  $q$  to  $N$ ,  $T$  and  $K$  that can be measured by the test vehicle. If the test car is travelling with the traffic stream, eq. (3) becomes

$$\frac{Nw}{Tw} = q - KV_w \dots\dots\dots(4)$$

If the test car is travelling against the travelling stream, eq. (3) becomes

$$\frac{Na}{Ta} = q - KV_a \dots\dots\dots (5)$$

Combining eq. (4) and eq. (5), gives

$$q = \frac{Na + Nw}{Ta + Tw} \dots\dots\dots (6)$$

To calculate the space mean speed  $u$  for the vehicle stream, from eq. (4),

$$\frac{Nw}{Tw} = q - \left(\frac{q}{u}\right)\left(\frac{L}{Tw}\right) \quad \text{Note that } q = Ku$$

Simplifying gives

$$\begin{aligned} \frac{Nw}{Tw} &= q - q\left(\frac{T}{Tw}\right) \quad \text{since } T = \frac{L}{u} \\ T &= T_w - \frac{Nw}{q} \dots\dots\dots(7) \end{aligned}$$

### EXAMPLE

Six runs were made along a two way highway between points A and B. flows were measured both with and against the cars moving. The following results were obtained.

Car moving from A to B

| Start | Stop  | No. of vehicles |           | met |
|-------|-------|-----------------|-----------|-----|
|       |       | overtaking      | overtaken |     |
| 16:05 | 16:16 | 2               | 1         | 401 |
| 16:34 | 16:44 | 3               | 2         | 360 |
| 17:05 | 17:17 | 4               | 1         | 419 |
| 17:35 | 17:44 | 5               | 3         | 397 |
| 18:05 | 18:18 | 2               | 1         | 407 |
| 18:35 | 18:45 | 2               | 3         | 412 |

Car moving from B to A

| start | Stop  | No. of vehicles |           | met |
|-------|-------|-----------------|-----------|-----|
|       |       | overtaking      | overtaken |     |
| 16:19 | 16:31 | 3               | 2         | 320 |
| 16:50 | 17:03 | 7               | 3         | 319 |
| 17:20 | 17:32 | 4               | 2         | 307 |
| 17:50 | 17:59 | 4               | 3         | 331 |
| 18:20 | 18:33 | 5               | 2         | 317 |
| 18:50 | 19:01 | 7               | 1         | 305 |

Distance from A to B is 6.4Km. Calculate the flow and stream speed in each direction.

### SOLUTION

Car traveling from A to B

| N <sub>a</sub> | N <sub>w</sub> | T <sub>a</sub> | T <sub>w</sub> | q<br>(veh./h) | T<br>(h) |
|----------------|----------------|----------------|----------------|---------------|----------|
| 320            | 1              | 12             | 11             | 837.39        | 0.18     |
| 319            | 1              | 13             | 10             | 834.78        | 0.17     |
| 307            | 3              | 12             | 12             | 775           | 0.20     |
| 331            | 2              | 9              | 9              | 1110          | 0.15     |
| 317            | 1              | 13             | 13             | 733.85        | 0.22     |
| 305            | -1             | 11             | 10             | 868.57        | 0.17     |
|                |                |                |                | 859.93        | 0.18     |

From eq. (6),

$$q = \frac{Na + Nw}{Ta + Tw}$$

$$q = \frac{320+1}{12+11} \times 60 = 837.39 \text{ veh./h}$$

From eq. (7)

$$T = Tw - \frac{Nw}{q}$$

$$T = 11 - \frac{1}{837.38}$$

$$T = 10.99 \text{ min}$$

$$= \frac{10.99}{60} = 0.18 \text{ h}$$

$$\text{Stream speed} = \frac{6.4}{0.18}$$

$$= 35.56 \text{ Km/h.}$$

Flow = 859.93 veh/h

Car moving from B to A

| $N_a$ | $N_w$ | $T_a$ | $T_w$ | $q$<br>(veh./h) | $T$<br>(h) |
|-------|-------|-------|-------|-----------------|------------|
| 401   | 1     | 11    | 12    | 1048.66         | 0.20       |
| 360   | 4     | 10    | 13    | 949.57          | 0.22       |
| 419   | 2     | 12    | 12    | 1052.5          | 0.20       |
| 397   | 1     | 9     | 9     | 1326.67         | 0.15       |
| 407   | 3     | 13    | 13    | 946.15          | 0.22       |
| 412   | 6     | 10    | 11    | 1194.27         | 0.18       |
|       |       |       |       | 1086.37         | 0.20       |

From eq. (6),

$$q = \frac{Na + Nw}{Ta + Tw}$$

$$q = \frac{401 + 1}{11 + 12} \times 60 = 1048.66 \text{ veh./h}$$

From eq. (7)

$$T = T_w - \frac{Nw}{q}$$

$$T = 12 - \frac{1}{1048.66}$$

$$= 12$$

$$= \frac{12}{60} = 0.20 \text{ h}$$

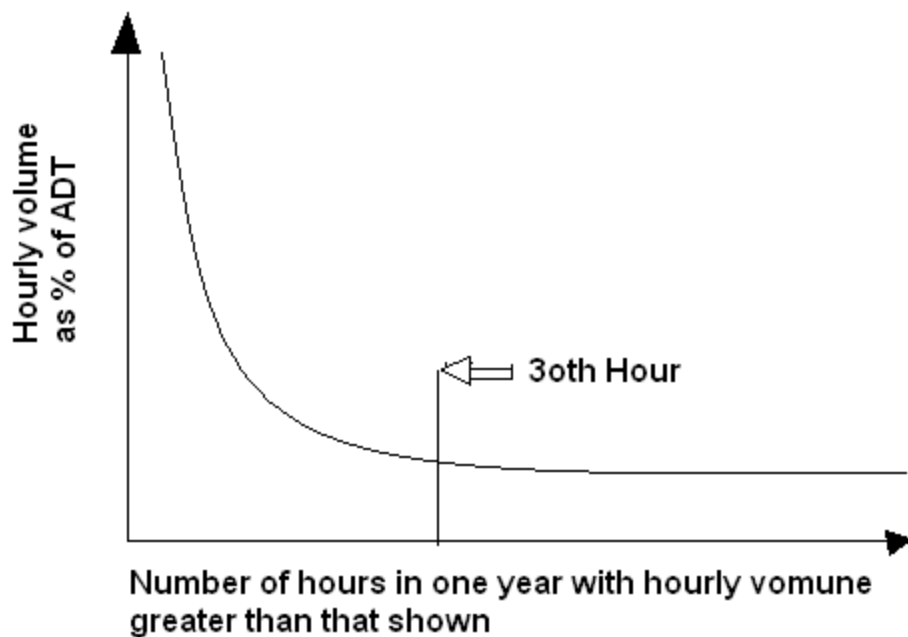
$$\text{Stream speed} = \frac{6.4}{0.2}$$

$$= 32 \text{ Km/h.}$$

Flow = 1086.37 veh./h

### DESIGN HOURLY VOLUME (DHV) DETERMINATION

The hourly volumes during the year are arranged in descending order of magnitude. They are converted to % of ADT. The hourly traffic as the % of ADT is plotted against number of hours in one with hourly volume greater than that shown.



DHV is then taken as the 30<sup>th</sup> highest hour expressed as a % of the ADT. Thus it is possible to measure and forecast volume in terms of ADT and multiply by a representative % to arrive at the DHV. This %, K, is typically 8 – 12 % for urban highways, 10 – 15 % for suburban highways and 12 – 18 % for rural highways.

### **ORIGIN AND DESTINATION (O AND D) SURVEYS**

The origin and destination (O and D) study is carried out mainly to

- Plan the road network and other facilities for vehicular traffic.
- Plan the schedule of different modes of transportation for the trip demand of commuters.

The O & D study of vehicular traffic determines their number, their origin and destination in each zone under study. The data may be supplemented by the number of passengers in each vehicle, purpose of each trip, intermediate stops made and reasons, etc. O & D study gives information like the actual direction of travel, selection of routes and length of trip. These studies are most essential in planning new highway facilities and in improving some of the existing ones. As an example there can be a high percentage of through traffic which may be diverted by providing a by-pass and considerable saving in distance and time can be made. This O & D study provides the basic data for determining the desired directions of flow or the desire lines. This is considered to be one of the important studies needed to solve many traffic problems in a zone and the most important study to plan the highway system in a region.

The various applications of O & D studies are

- To judge the adequacy of existing routes and to use the study in planning new network of roads.
- To plan transportation system and mass transit facilities in cities including routes and schedules of operation.
- To locate express ways or major routes along the desire lines.
- To establish preferential routes for various categories of vehicles including by-pass.
- To locate terminals and to plan terminal facilities.
- To locate new bridges as per traffic demand.
- To locate intermediate stops of public transport.
- To establish design standards for the roads, bridges and culverts along the route.

## **METHODS OF COLLECTING O & D DATA**

There are five commonly adopted methods for collecting O & D data. The choice of the method depends on the objective and location.

### **1. ROAD SIDE INTERVIEW METHOD**

The vehicles are stopped at previously determined interview stations by a group of persons and the answers to prescribed questionnaire are collected on the spot. The information collected include the place and time of origin and destination, route, location of stoppages, the purpose of the trip, type of vehicle and number of passengers in each vehicle. The traffic may be filtered through a prescribed lane by previous warning signs and with the help of police so that each driver of the selected sample of vehicles is interviewed. The % of sample interviewed out the total traffic in each selected period should be noted from appropriate traffic volume study taken simultaneously.

In this method, the data can be collected quickly in a short duration and the field organization is simple and the team can be trained quickly. The main drawback is that the vehicles are stopped for interview and there is delay to the vehicular movement. Also road users are always unwilling to stop and also unless there is enough space, undue congestion may result due to stopped vehicles.

### **2. LICENSE PLATE METHOD**

The entire area under study is mapped out and the observers are simultaneously stationed at all points of entry and exit on all the routes leading to and out of the area. Each observer at the observation station is given synchronized time pieces and they note the license plate number (registration number) of the vehicle entering and leaving the mapped area and the time. Separate recording sheets are maintained for each direction of movement for a specified time interval.

After collecting the field data, the computation and analysis of the data is carried out by tracking each vehicle and its time of entering and leaving the mapped area.

The merits of this method are that it is quite easy and quick as far as the field work is concern and the field organization can be trained quickly. However, its demerits are that there is a lot of computation in tracing the trip through a network of stations and unless there is a large network of stations to take observation along the route of the vehicle, it is not easy to get the information of the routes followed by the vehicle. Hence, a large number of things are required to take simultaneous observation when a large area is to be surveyed. However, this method is quite advantageous when the area of study is considerably small like a large intersection or a small business centre.

### **3. RETURN POST CARD METHOD**

In this method prepaid business reply postcards with return address are distributed to the road users at some selected points along the route or the cards are mailed to owners of vehicles. The questionnaire to be filled in by the road users is printed on the card, along with a request for co-operation and purpose of study. The distributing stations for the cards may be selected where the vehicles have to stop as in the case of a toll gate.

This method is suitable when the traffic is heavy. The personnel need not be skilled or just trained for distributing the cards. Only a part of the road users may return the cards promptly after filling in the desired details properly and correctly. If conclusions are drawn in such cases, it is likely that these may not give a true picture.

### **4. TAG ON CAR METHOD**

In this method a pre-coded card is attached on a vehicle as it enters the area of study. When the car is leaving the study area, the other observations are recorded on the tag. This method is useful when the traffic is heavy and moves continuously. But this method gives only information regarding the point of entering and exit and the time taken to traverse the study area.

### **5. HOME INTERVIEW METHOD**

In this method, a random sample of 0.5 – 10 % of the population is selected and their residences are visited by the trained personnel who collect the travel data from each member of the household. Detailed information regarding the trips made by the members is obtained on the spot. The problem of stopping vehicles and consequent difficulties are avoided altogether. The present traffic needs are clearly known and the analysis is also simple. Additional data including socio-economic and other details may be collected so as to be useful for forecasting traffic and transportation growth.

## **HIGHWAY GEOMETRICS**

Geometric design of highway refers to the physical proportioning of the road way as distinguished from other types of design such structural design i.e. geometric design is primarily concern with relating the physical elements of the highway to the requirements of the driver and the vehicle. It is mainly concern with those elements which make up the visible features of the road way and does not include the structural design of the facilities.

Proper geometric design will inevitably reduce the number and severity of highway accidents while ensuring high traffic capacity with the minimum of delay to vehicles. The basic components of high way design are

- The cross section
- Horizontal alignment
- Vertical alignment
- Intersection element
- Sight distance consideration

## **FACTORS THAT AFFECT GEOMETRIC DESIGN**

The general factors which directly affect the geometric design are

- ✓ Design speed
- ✓ Topography
- ✓ Traffic factors
- ✓ Design hourly volume and capacity
- ✓ Environmental and other factors.

The major factors which should normally engage serious attention of the highway engineer are those which affect horizontal alignment, vertical alignment and the cross section.

## **DESIGN SPEED**

- Design speed (DS) is defined as a selected speed to determine the various geometric features of the roadway
- DS is the highest continuous speed @ which driver can travel safely during good weather condition
- A design speed is selected to achieve a desired level of operation and safety on the highway . It is one of the first parameters selected in the design process because of its influence on other design variables
- DS depends on the functional classification of the highway, the topography of the area in which the highway is located, & the land use of the adjacent area
- Drivers travel at different speeds depending on their behaviour. For the purpose geometric design, a single representative speed, known as design speed is required

- In selection of DS, effort should be made to attain a desired combination of safety, mobility, and efficiency within the constraints of environmental quality, economics, aesthetics, and social or political impacts
- Once the DS is selected, all of the pertinent highway features should be related to it to obtain a balanced design
- The selected DS should be a logical one with respect to the anticipated operating speed, topography, the adjacent land use, & the functional classification of the highway
- It is selected from standards for designing specific section of road considering the terrain, land use, classification, & so forth
- Based on a road functional class, terrain & design traffic, a design class with respect to DS is chosen from standard to the design parameters associated with that class & the required geometric constraints are fixed

### TOPOGRAPHY

- The geometric design parameters of a road depend on the traverse terrain through which the road passes
- Terrain is basically categorized into level (flat), rolling & mountainous

**Level terrain** → where segments contain flat grades of  $\leq 2\%$  such that heavy vehicles are able to maintain the same speed as passenger cars throughout the segment



**Rolling terrain** → where segments contain short or medium length grades of  $\leq 4\%$ . Heavy truck speeds are lower than passenger cars but are not at crawl speed



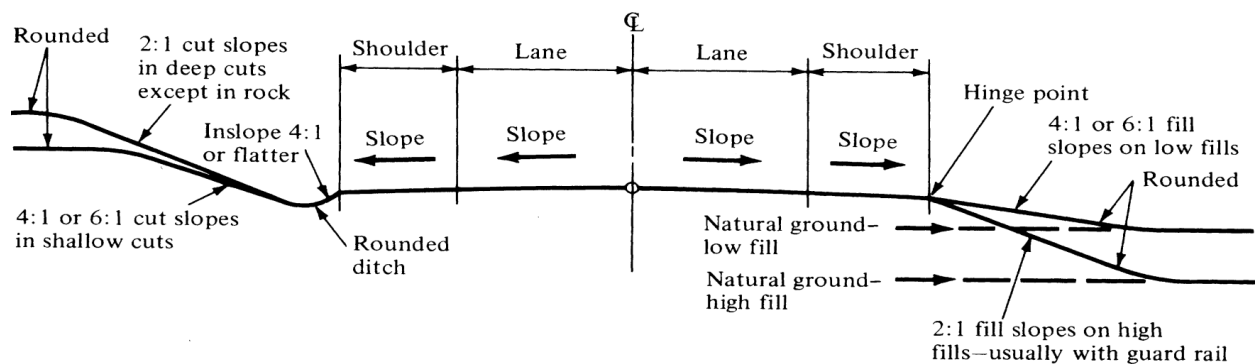
**Mountainous terrain** → where segments contain grades of  $> 4\%$  of about 1.0 km or more. Heavy vehicles speeds are much lower than passenger cars, specifically, travelling at crawl speed.

A passing lane is usually added within such section in level or rolling terrain or as a truck climbing lane



### HIGHWAY CROSS-SECTION ELEMENTS

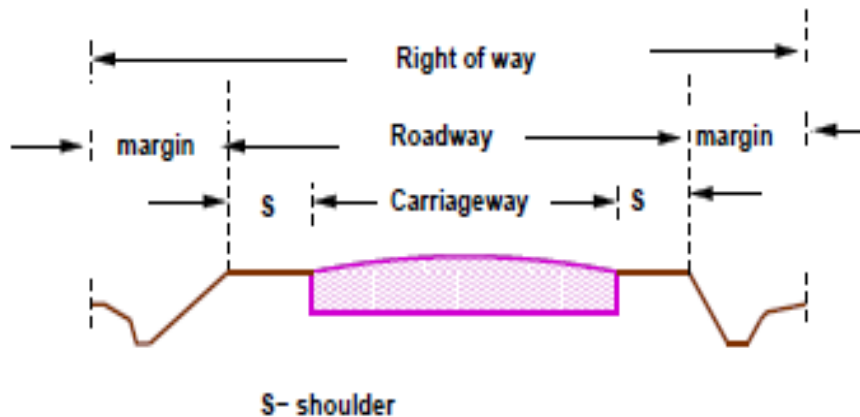
- Highway cross section defines the configuration of roadway perpendicular to the centre-line
- The key elements of a highway cross section comprise of the travel lanes, shoulders, & medians (for some multilane highways)
- Other elements include roadside barriers, curbs, gutters, guard rails, sidewalks, & side slopes



**Two-lane Highway Cross-section**

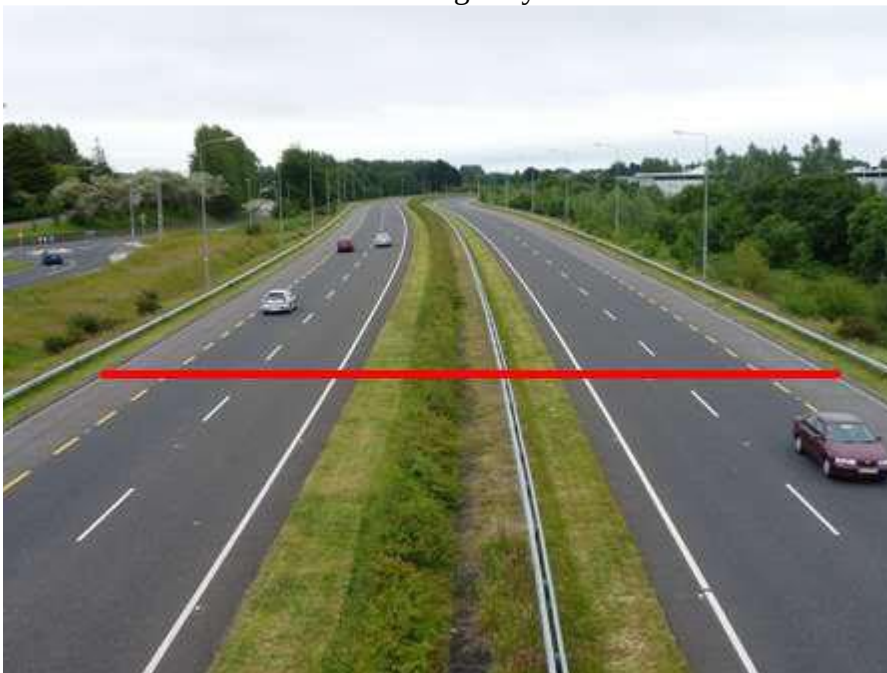
### RIGHT OF WAY (ROW)

- It is the width of land secured & preserved to the public for road purposes
- ROW should be adequate enough to accommodate all the elements that constitute the cross section of the roadway & could reasonably allow provision for future development
- It is the distance between boundary stones of road on either side of road



### FORMATION WIDTH

- Formation width is the top width of the highway embankment or the bottom width of cutting excluding the side drain
- Formation width = Width of Carriageway + Width of shoulder



### 1. CARRIAGE WAY

This is the part of the road way which is designed to accommodate traffic. The carriage way may be single lane, two lanes or more. It can be divided (dual) or undivided.



## **2. CAMBER (CROSS SLOPE)**

The camber or cross slope is a slope provided in the pavement cross section to facilitate the drainage of rain water from the road surface. The pavement is raised at the middle (crown) and slopes downward toward both edges for most linked roads or wider undivided roads. The amount of camber or cross slope depends upon the intensity of rainfall and the smoothness of the surface. Drainage and quick disposal of water from the pavement surface by providing cross slope is considered important because of the following reasons.

- To prevent the entry of water into the sub grade soil through the pavement layers. The stability, surface condition and the life of the pavement get adversely affected if water enters in the sub grade and the soil gets soaked.
- To prevent the entry of water into the bituminous pavement layers, as continues contact with water causes stripping of the bitumen from the aggregates and results in deterioration of the pavement layer.
- To remove the water from the pavement surface as quickly as possible and to allow the pavement to get dry soon after the rain so as not to decrease the skid resistance of the road thereby rendering it slippery and unsafe for vehicles operation at high speeds.

## **3. SHOULDERS**

This is the portion of the roads way which is intended for the accommodation of stopped vehicles, emergency use, pedestrians, etc.

## **4. CURBS OR KERBS**

They are provided to serve the following functions:

- For aesthetics

- For protection and strengthening of pavement edge.
- For the control and ease of drainage.

## 5. **MEDIAN**

The median is provided to prevent head-on collision between vehicles moving in opposite direction on adjacent lanes. Apart from this, it is provided for:

- ❖ Drainage facility
- ❖ Vehicles storage space for crossing or left turn movement at intersection
- ❖ Aesthetic treatment including painting to reduce high light glare for opposing vehicles
- ❖ Safety area for out of control or disabled vehicles.

## 6. **Sidewalks**

- ❖ Sidewalks are usually provided along roadways in urban areas for pedestrians use
- ❖ Nevertheless, the provision of sidewalks in rural areas should be evaluated & the planning process to determine sections of the road where they are needed
- ❖ E.g., along rural principal arterials may require sidewalks in areas with high pedestrian concentrations, such as adjacent to schools, industrial plants, & local businesses
- ❖ Generally, sidewalks should be provided when pedestrian traffic is high along main or high-speed roads in either rural or urban areas
- ❖ When shoulders are not provided on arterials, sidewalks are necessary even when pedestrian traffic is low
- ❖ In urban areas, sidewalks should also be provided along both sides of collector that serve as pedestrian access to schools, parks, shopping centers, transit stops, & in commercial areas
- ❖ Sidewalks should have a minimum clear width of 4 ft in residential areas & a range of 4 to 8 ft in commercial areas
- ❖ To encourage pedestrians to use sidewalks, they should have all-weather surfaces, otherwise, pedestrians will tend to use traffic lanes rather than unpaved or slippery sidewalks

## 7. **Guard Rails**

- Guard rails are longitudinal barriers placed on the outside of sharp curves & at sections with high fills
- They serve to prevent vehicles from leaving the roadway
- They are installed at embankments higher than 8 ft and when shoulder slopes are greater than 4:1
- Shapes commonly used include the W beam & the box beam
- The weak post system provides for the post to collapse on impact, with the rail deflecting & absorbing the energy due to impact

### **HIGHWAY ALIGNMENT**

An ideal and most interesting highway is one that follows the existing natural topography of the country. This is the most economical to construct, but there are certain aspects of design that must be adhered to, that prevent the designer from following this undulating surface without making certain adjustment in the vertical and horizontal direction.

The designer should provide alignments in which conditions are consistent. Sudden changes in alignment should be avoided as much as possible e.g. long tangent should be connected with long curve.

### **HORIZONTAL ALIGNMENT**

The horizontal alignment represents the projection of the facilities on the horizontal plane. It generally consists of straight line segments known as tangents connected by circular curves either directly (simple curves) or through intermediate transition curve. The length of the facility is measured from a control line (such as center of the road way) and is usually expressed in terms of 100m from reference point. Many design factors are to be considered in the horizontal alignment. These include design speed, radius of circular curves, type and length of transition curves, super elevation and widening of pavement on curves.

The alignment should enable consistent, safe and smooth movement of the vehicles operating at design speed. Hence, it is necessary to avoid those sharp curves and reverse curves which could not be conveniently negotiated by the vehicles at design speed. Improper design of horizontal alignment of roads would necessitate speed changes resulting in increased vehicle operation cost and accident rate.





### **Considerations in Horizontal Alignment Design**

Horizontal alignment is expected to meet the following general considerations:

- Alignment should be straight if possible within physical & economic constraints

Alignment should be consistent

- Try to avoid sharp curves at the ends of long tangents and sudden changes from gently to sharply curving alignment.

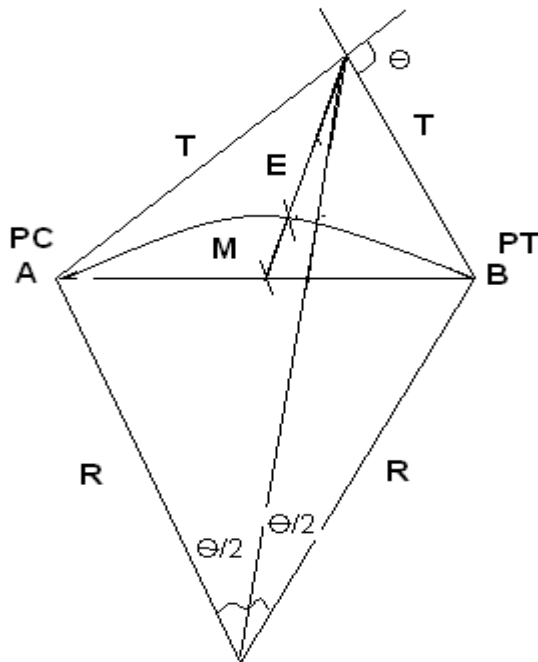
- Avoid the use of minimum radii whenever possible
- Avoid horizontal curves on bridges
- On major roads and freeways, the minimum curve length in meters should be three times the design speed in km/h

### PROPERTIES OF CIRCULAR CURVES

A simple circular curve connects two tangents which when projected meets at a point of intersection symbolized as PI. Preceding a direction of the increasing station value, a point A is designed as the point of curvature symbolized as PC, the beginning of the curve and point B at the end of the curve denoted as point of tangency PT. Two straight lines intersecting at PI can be connected with infinite number of circular curves. Each curve can be defined by its radius or degree of curvature 'D' for which two alternative definitions are encountered in practice (the cord or arc definitions). The arc definition is preferred by the highway engineer and is defined as the central angle in degree subtended by an arc of length 100m. The circumference of a circle bears the relationship to 360° as 100m arc does to the degree curve D.

$$\text{Therefore } \frac{2\pi R}{360} = \frac{100}{D}$$

$$= DR = \frac{18000}{\pi} \text{ where } D = \text{degree}$$



T= Tangent length

PI= Point of intersection

E= External distance

PC= Point of curvature (A)

M= Middle ordinate distance (Chord to curve distance)

L<sub>c</sub>=Long chord (A to B)

PT= Point of tangency (B)

L= Arc length

Θ= Deflection angle or external angle.

$$1 \ L = \frac{\theta}{360} \times 2\pi R$$

$$2 \ T = R \tan \left( \frac{\theta}{2} \right)$$

$$3 \ L_c = 2R \sin \left( \left( \frac{\theta}{2} \right) \right)$$

$$4 \ M = R - R \cos \left( \left( \frac{\theta}{2} \right) \right)$$

$$5 \ E + M = T \sin \left( \left( \frac{\theta}{2} \right) \right)$$

$$E = T \sin \left( \left( \frac{\theta}{2} \right) \right) - M$$

$$E = R \left( \sec \left( \left( \frac{\theta}{2} \right) \right) - 1 \right)$$

### EXAMPLE

A circular curve of a highway have to be set out to pass through a point 40m from the intersection point and equidistance from the tangent points. The straights are deflection through 46°40'. Calculate the radius of the curve and the tangent length.

### SOLUTION

Given

$$E = 40, \Theta = 46^\circ 40', R = ? \ T = ?$$

From the equation  $E = R \left( \sec \left( \left( \frac{\theta}{2} \right) \right) - 1 \right)$

$$R = E / \left( \sec \left( \left( \frac{\theta}{2} \right) \right) - 1 \right)$$

$$= 40 / \left( \sec(46^\circ 40' / 2) - 1 \right)$$

$$= 449.1\text{m}$$

$$\text{Then } T = R \tan \left( \left( \frac{\theta}{2} \right) \right)$$

$$= 449.1 \tan (46^\circ 40' / 2)$$

$$= 193.7\text{m}$$

### DESIGN SPEED

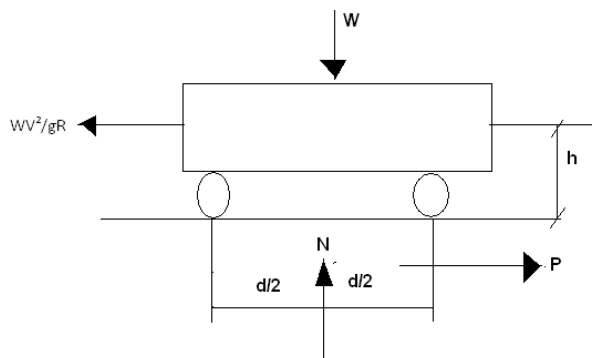
The design speed is the main factor on which geometric design of elements depend. The sight distances, extra width of pavement, length of horizontal curve, radius of horizontal transitional

curves, super elevation and length of vertical curves are dependent on the design speed. The choice of the design speed depends on

- Class of road (urban, rural, suburban, arterial, etc)
- Terrain (plain, rolling, mountainous or steep)
- Traffic volume
- Economic consideration.

### CURVE AND CENTRIFUGAL FORCE

Anybody moving rapidly about a curved path is subjected to an outward reactive force called the centrifugal force. On a highway curve, this force tends to cause vehicles to overturn or slide outside from the centre of the road curvature.



The figure illustrates forces acting on the vehicle as it moves about a horizontal curve. Since the carriage way is flat and forces are in equilibrium, then

$$P = Wv^2/gR$$

where  $W$  = weight of vehicle in Kg.

$V$  = speed of vehicle in m/s

$R$  = radius of curve in m

$g$  = acceleration due to gravity

$P$  = lateral frictional force resisting the centrifugal force.

$$\text{From } P = \mu N = \mu W$$

$$= \mu W = Wv^2/gR$$

$$= \mu = v^2/gR = P/W$$

where  $\mu$  is the developed lateral coefficient of friction.

$P/W = v^2/gR$  is known as the centrifugal ratio.

This centrifugal force also tends to cause a vehicle to overturn. This force acts through the centre of gravity of the vehicle causing an overturning moment about the point of contact between the outer wheels of the vehicle and the carriage way. This overturning moment caused by this centrifugal force is resisted by the weight of the vehicle acting through its centre of gravity. Thus for an equilibrium position,

$$WV^2/gR \times h = W \times d/2$$

$$h = d / (2V^2/gR) = d/2\mu$$

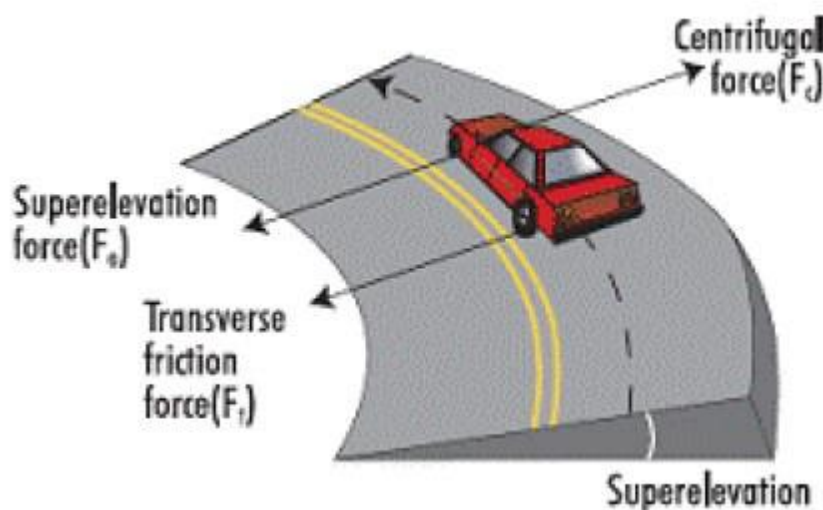
where  $d$  is the lateral distance between the wheels.

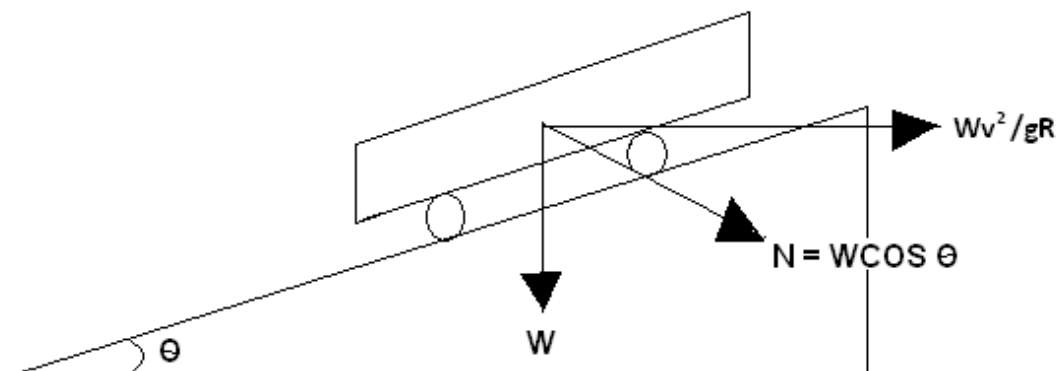
$h$  is the height of the centre of gravity above the carriage way.

The above equation illustrates that for instance if  $\mu = 0.5$ , the height of centre of gravity must be greater than the lateral distance before overturning will take place. Modern vehicles however, and passenger cars have low centres of gravity so that relatively high values of  $\mu$  have to be developed before overturning could take place. In practice, the frictional value is sufficiently low for sliding to take place long before overturning.

### SUPER ELEVATION

In order to counteract the effect of centrifugal force and to reduce the tendency of the vehicle to overturn or skid, the outer edge of the pavement is raised with respect to the inner edge, thus providing a transverse slope throughout the length of the horizontal curve. This transverse inclination to the pavement surface is known as super elevation or cant or banking. Thus, super elevation can be defined as the transverse inclination given to the cross section of the road surface throughout the length of the horizontal curve to reduce the effect of centrifugal force on the running wheels.





For equilibrium,  $\Sigma F = 0$

$W \sin \theta + f W \cos \theta = (W V^2 \cos \theta) / g R$  where  $f$  = frictional coefficient.

Dividing through by  $W \cos \theta$ ,

$$\tan \theta + f = V^2 / g R$$

Let  $\tan \theta = e$  = super elevation

$$e + f = V^2 / g R$$

$$e = (V^2 / g R) - f$$

where  $e$  = rate of super elevation

$f$  = design value of lateral friction coefficient

$R$  = radius of curve in m

$V$  = speed of the vehicle, in m/s

$G$  = acceleration due to gravity =  $9.8 \text{ m/s}^2$

If the speed of the vehicle is in km/h, the above equation becomes

$$e = (V^2/127R) - f$$

#### **ADVANTAGES OF SUPER ELEVATION**

- Higher speed without any danger of overturning
- There is increase in the volume of traffic
- Maintenance is reduced; otherwise, there will be wearing on the outer wheels causing potholes.
- Water can be drained easily.

#### **EXAMPLES**

1. A horizontal curve is to be designed in Federal Polytechnic Mubi campus road. Calculate

(a) Desirable

(b) Absolute

Minimum radius for super elevation assuming

(i) Desirable design speed = 90Km/h

(ii) Minimum design speed = 40Km/h

(iii) Maximum super elevation  $e = 7\%$

(iv) Frictional coefficient  $f = 0.14$

2. The radius of a horizontal curve is 100m. The design speed is 50Km/h and the coefficient of lateral friction is 0.15

(a) Calculate the super elevation required if full lateral friction is assumed to be developed

(b) Calculate the coefficient of friction needed if no super elevation is provided.

(c) Calculate the equilibrium super elevation if the pressure on the inner and outer wheels should be the same.

#### **ATTAINMENT OF SUPER ELEVATION**

The attainment of super elevation requires that the carriage way cross section should be tilted on the transition or circular curve. The tilting of the carriage way may be carried out by one of the following methods:

1. Rotation about the centre line of the carriage way: This method is usually accomplished by lowering the inside edge of the pavement and raising the outside edge without changing the centre line grade.
2. Rotation about the inner edge: Here the inner edge retains its normal grade but the centre line and the outer edge are raised.
3. Rotation about the outer edge: In contrast to rotation about the inner edge, the centre line and the inner edge grades are lowered.

Regardless of which method is used, care should always be taken so that the carriage way is not affected adversely by the rotation.

### **MAXIMUM SUPER ELEVATION**

As per equations of super elevation, the value of super elevation needed increases with increase in speed and with decrease in radius of the curve, for a constant value of coefficient of lateral friction  $f$ . From the practical point of view, it will be necessary to limit the maximum allowable super elevation to avoid very high value of  $e$ . This is particularly necessary when the road has to cater for mixed traffic, consisting of fast and slow traffic.

In the case of heavily loaded bullock carts and trucks carrying less dense materials like cotton, the centre of gravity of the loaded vehicle will be very relatively high and it will not be safe for such vehicles to move on the road with a high rate of super elevation. Because of the slow speed, the centrifugal force will be negligibly small in the case of bullock carts. Hence, to avoid the danger of toppling of such loaded slow moving vehicles, it is essential to limit the value of maximum allowable super elevation.

### **MINIMUM SUPER ELEVATION**

From the drainage considerations it is necessary to have a minimum cross slope to drain off the surface water. If the calculated super elevation from  $e = (V^2/gR) - f$  works out to be equal to or less than the camber of the road surface, then the minimum super elevation to be provided on the horizontal curve may be limited to the camber of the surface. Thus, after the elimination of the crown a uniform cross slope equal to the camber is maintained from outer to inner edge of pavement at the circular curve.

### **SIGHT DISTANCE**

The ability to see ahead is of most importance to the safe and efficient operation of a highway i.e. highways must be designed to give drivers a sufficient distance of clear vision ahead so that they can avoid unexpected obstacles and can pass slower vehicles without danger.

Sight distance is the length of the highway visible ahead to the driver of a vehicle. Highway designers are concerned with two types of sight distances.

1. **OVERTAKING SIGHT DISTANCE:** This is the minimum distance needed by a driver to overtake another vehicle safely and comfortably without interfering with the speed of an oncoming vehicle travelling at the design speed should this vehicle come into view after the overtaking manoeuvre has started.
2. **STOPPING SIGHT DISTANCE:** It is the distance needed to stop a vehicle moving at the design speed of the road if an unexpected object is sighted on the roadway. For safety, a minimum stopping sight distance is required on multilane highways and freeways as well as for two lane roads. Minimum stopping sight distance is based on the sum of two distances:

- ✓ Distance travelled from the time the object is sighted to the time the brakes are applied which depends on
  - The speed of the vehicle
  - The perception time of the driver
  - Brake reaction time of the driver
- ✓ Distance required for stopping the vehicle after the brakes have been applied which depends on
  - Speed of the vehicle
  - Conditions of the brakes, type and roadway surface
  - The alignment and grade of the highway

### ANALYSIS OF STOPPING SIGHT DISTANCE (SSD)

The stopping sight distance is the sum of lag distance and the braking distance i.e.

SSD = Lag distance + Braking distance

**LAG DISTANCE:** During the time when the object is sighted to the time the brakes are applied, the vehicle may be assumed to proceed forward with a uniform speed. If  $V$  is speed in m/s and  $t$  is the total reaction time of the driver in s, the lag distance is

$L = Vt$  (m) but if the speed  $V$  is in Km/h, then

$L = 0.278Vt$  (m)

**BRAKING DISTANCE:** This is the distance travelled by the vehicle after the application of the brakes to a stopped position and is given by

$L = V^2/2gf$  but if  $V$  is in Km/h, then

$L = V^2/254f$

**SSD =  $0.278Vt + V^2/254(f \pm G)$**

Where  $V$  = speed of vehicle in Km/h

$t$  = total reaction time of the driver

$f$  = coefficient of friction

$g$  = grade of highway which is negative when vehicle is descending and positive when vehicle is ascending.

### EXAMPLE

Calculate the SSD on a descending gradient of 2% for a design speed of 80Km/h assuming that the total reaction time is 2.5s,  $f = 0.35$

### SOLUTION

SSD =  $0.278Vt + V^2/254(f \pm G)$

=  $0.278 \times 80 \times 2.5 + 80^2/254(0.35 - 0.02)$

$$= 55.6 + 76.35$$

$$= 131.95\text{m}$$

**NOTE: READ ON PAVEMENT WIDENING**

### **DETERMINATION OF THE NUMBER OF LANES**

The number of lanes should be capable of accommodating the anticipated number of vehicles. The highway capacity manual recommends a simple analysis to determine the probable number of lanes required and where a multiple lanes is appropriate for the expected conditions.

The planning procedures require information on:

- The general terrain through which the highway is to be viewed
- The annual average daily traffic (AADT) for the designing year
- The % of the traffic in the peak direction of flow symbolizes as D
- The peak hour factor (PHF) for the designing year
- The % of trucks in the traffic strength, T
- The type of multilane highway and its development environment procedure

**Step 1:** Convert the annual average daily traffic (AADT) to directional design hourly volume (DDHV) as follows

$$\text{DDHV} = \text{AADT} \times K \times D$$

Where K = % of AADT that occur at peak hour

D = directional factor

**Step 2:** Specify service flow rate  $\text{SFL}_i$  from the table using % of trucks and level of service.

**Step 3:** Estimate number of lanes as follows

$$N = \text{DDHV} / (\text{SFL}_i \times F_E \times \text{PHF})$$

Where  $F_E$  = adjustment factor for the type of highway

PHF = Peak Hour Factor, defined as the ratio of the total hourly volume to the maximum 15 minutes rate of flow within the hour.  $\text{PHF} = V / (4 \times V_{15})$

### **EXAMPLE 1**

An undivided highway is to be constructed within a suburban environment with a rolling terrain. The AADT = 20,000veh/day with 10% truck. The fraction of traffic in the peak direction of flow is 0.6, the peak hour factor (PHF) is 0.95 and the desired level of service is B. Assuming that K = 0.15, determine the number of lanes that will be needed in each direction.

### **SOLUTION**

$$\text{AADT} = 20,000 \text{ veh/day}$$

$$D = 0.6, K = 0.15, \text{ level of service} = B, \text{ PHF} = 0.95, T = 10\%$$

From the tables,

$$F_E = 0.80$$

$$\text{SFL}_B = 850$$

$$\begin{aligned}
 DDHV &= AADT \times K \times D \\
 &= 20,000 \times 0.6 \times 0.15 \\
 &= 1800 \text{ veh/day} \\
 N &= DDHV / (SFL_B \times F_E \times PHF) \\
 &= 1800 / (850 \times 0.8 \times 0.95) \\
 &= 2.8 \\
 N &= 3 \text{ lanes per direction}
 \end{aligned}$$

### CLASS ASSIGNMENT

Given the following data:

1. Maximum 15 minutes rate of rate of flow within an hour = 9000veh.
2. P.H.F = 0.85
3. K = 0.15
4. Fraction of traffic in peak direction = 0.60
5.  $SFL_i = 1750$
6.  $F_E = 0.8$

Determine the number of lanes this particular highway should have assuming that the total number of vehicles in that hour is the AADT for the designing year.

### EXAMPLE 2

- a. A divided highway is to be constructed in a rural environment with a plain terrain. The forecast AADT is 15000veh/day. No truck is allowed to use this highway. The fraction of traffic in the peak direction of flow is 0.65, the peak hour factor is 0.90 and the desired level of service is A. Assuming  $k = 0.15$ , determine the number of lanes that will be needed to accommodate the anticipating traffic in each direction.

### $SFL_i$ TABLE

#### LEVEL TERRAIN

| level of service | % OF TRUCK |      |      |      |      |      |      |      |
|------------------|------------|------|------|------|------|------|------|------|
|                  | 0          | 2    | 4    | 6    | 8    | 10   | 15   | 20   |
| A                | 700        | 700  | 700  | 650  | 650  | 650  | 650  | 600  |
| B                | 1100       | 1100 | 1050 | 1050 | 1050 | 1000 | 1000 | 1000 |
| C                | 1400       | 1400 | 1350 | 1350 | 1350 | 1350 | 1250 | 1250 |
| D                | 1750       | 1750 | 1700 | 1650 | 1650 | 1650 | 1650 | 1550 |
| E                | 2000       | 2000 | 1950 | 1800 | 1800 | 1850 | 1800 | 1750 |

**ROLLING TERRAIN**

| level of service | % OF TRUCK |      |      |      |      |      |      |      |
|------------------|------------|------|------|------|------|------|------|------|
|                  | 0          | 2    | 4    | 6    | 8    | 10   | 15   | 20   |
| A                | 700        | 650  | 600  | 600  | 550  | 550  | 500  | 500  |
| B                | 1100       | 1050 | 1000 | 950  | 900  | 850  | 800  | 700  |
| C                | 1400       | 1300 | 1250 | 1200 | 1150 | 1100 | 1000 | 900  |
| D                | 1750       | 1650 | 1550 | 1500 | 1400 | 1350 | 1250 | 1100 |
| E                | 2000       | 1900 | 1800 | 1700 | 1550 | 1550 | 1450 | 1250 |

**MOUNTAINOUS TERRAIN**

| level of service | % OF TRUCK |      |      |      |      |      |      |     |
|------------------|------------|------|------|------|------|------|------|-----|
|                  | 0          | 2    | 4    | 6    | 8    | 10   | 15   | 20  |
| A                | 700        | 650  | 600  | 500  | 450  | 400  | 350  | 300 |
| B                | 1100       | 950  | 850  | 700  | 700  | 650  | 550  | 450 |
| C                | 1400       | 1250 | 1100 | 1000 | 900  | 850  | 700  | 600 |
| D                | 1750       | 1500 | 1350 | 1250 | 1100 | 1050 | 850  | 750 |
| E                | 2000       | 1750 | 1550 | 1400 | 1250 | 1200 | 1000 | 950 |

 $F_E$ 

| TYPE     | DIVIDED | UNDIVIDED |
|----------|---------|-----------|
| SUBURBAN | 0.9     | 0.8       |
| URBAN    | 1       | 0.95      |

**INTERSECTIONS**

At an intersection, there are through, turning and crossing traffic and those traffic movements may be handled in different ways depending on the type of intersection and its design. Generally, intersection problems are unavoidable except cases of expressway systems where such problems

are avoided by providing grade separated intersections and controlled areas. The efficiency, safety, speed, cost of operation and capacity of road system depends very much on the intersection design. Intersections may be classified into two broad groups:

1. **INTERSECTION AT GRADE:** These include all roads that meet at more or less the same level. The traffic manoeuvres like merging, diverging and crossing are involved in the intersection at grade.

2. **GRADE SEPARATED INTERSECTION (INTERCHANGES):** The intersecting roads are separated by difference in level, thus eliminating the crossing manoeuvres.

Some of the factors to be considered in intersection design include relative speed and manoeuvre area.

### **INTERSECTION AT GRADE**

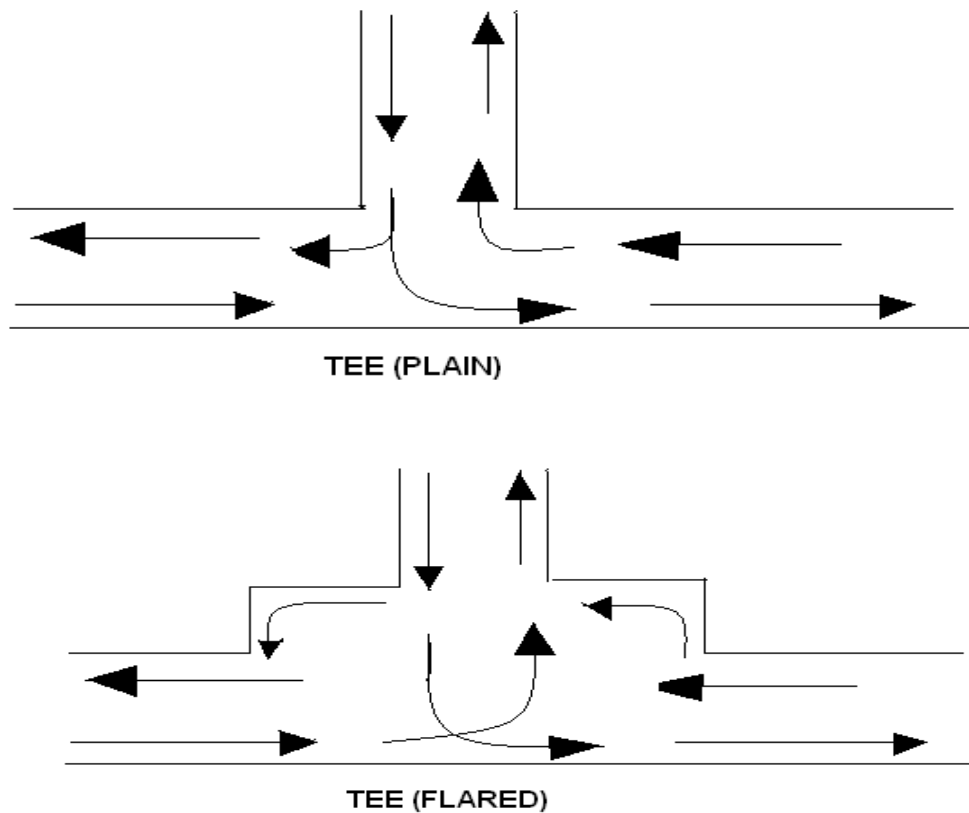
All road intersections which meet at about the same level allowing traffic manoeuvres like merging, diverging and weaving are called intersections at grade. These intersections can further be divided into channelized, unchannelized and rotary intersections.

### **BASIC REQUIREMENTS OF INTERSECTION AT GRADE**

- At the intersection the area of conflict should be as small as possible.
- The relative speed and particularly the angle of approach of vehicle should be small.
- Adequate visibility should be available for vehicles approaching the intersection.
- Sudden change of path should be avoided.
- Geometric features like turning radius and width of pavement should be adequately provided
- Proper signs should be provided on the roads approaching intersection to warn drivers.
- Good lighting at night is desirable.
- If the number of pedestrians and cyclists are large, separate provision should be made for their safe passage in intersections with high volume of fast moving traffic.

### **1. UNCHANNELIZED INTERSECTION**

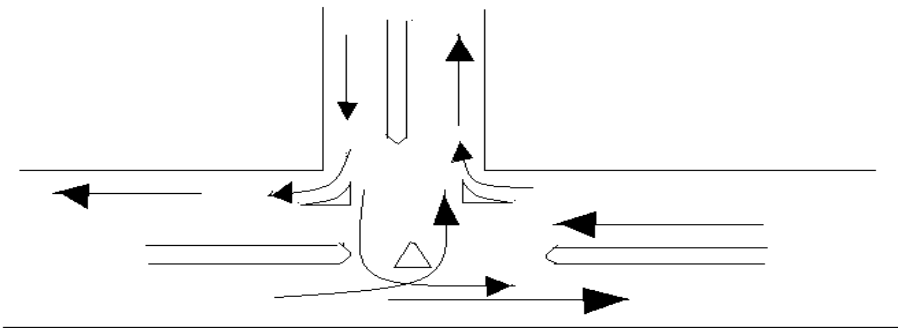
The intersection area is paved and there is absolutely no restriction to vehicles to use any part of the intersection area. Hence, these types of intersection are the lowest class of intersection, easiest in design but most complex in traffic operations resulting in maximum conflict area and more number of accidents unless controlled by traffic signs or police. When no additional pavement width for turning is provided, it is called a plain intersection but when the pavement is widened at the intersection area by traffic lane or more, it is known as flared intersection.



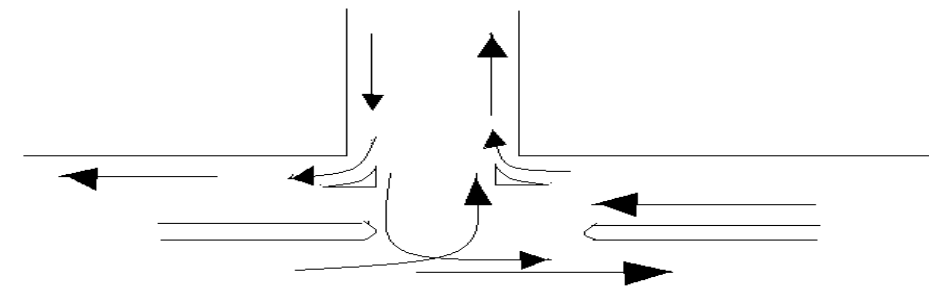
The arrows indicate the path of traffic flow, turning, crossing and through movements. It may be seen that the conflict area is quite large as path of turning vehicles are not restricted or controlled. One of the crossing vehicles will have to stop while the other proceeds.

## 2. CHANNELIZED INTERSECTIONS

Channelized intersection is achieved by introducing islands into the intersection area, thus reducing the total conflict area available in the unchannelized intersection. The radius of the entrance and exit curves and the area are suitably designed to accommodate the channelizing island of proper size and shape. These islands help to channelize turning traffic, to control their speed and angle of approach and to decrease the conflict area at the intersection. Channelization may either be partial or complete with directional islands and medians. From the traffic operation point of view, there is better control of traffic entering and leaving the intersection and hence, channelized intersections are considered superior to the all-paved types. However, one of the crossing vehicles will have to stop while the other proceeds.



TEE (COMPLETE CHANNELIZATION)



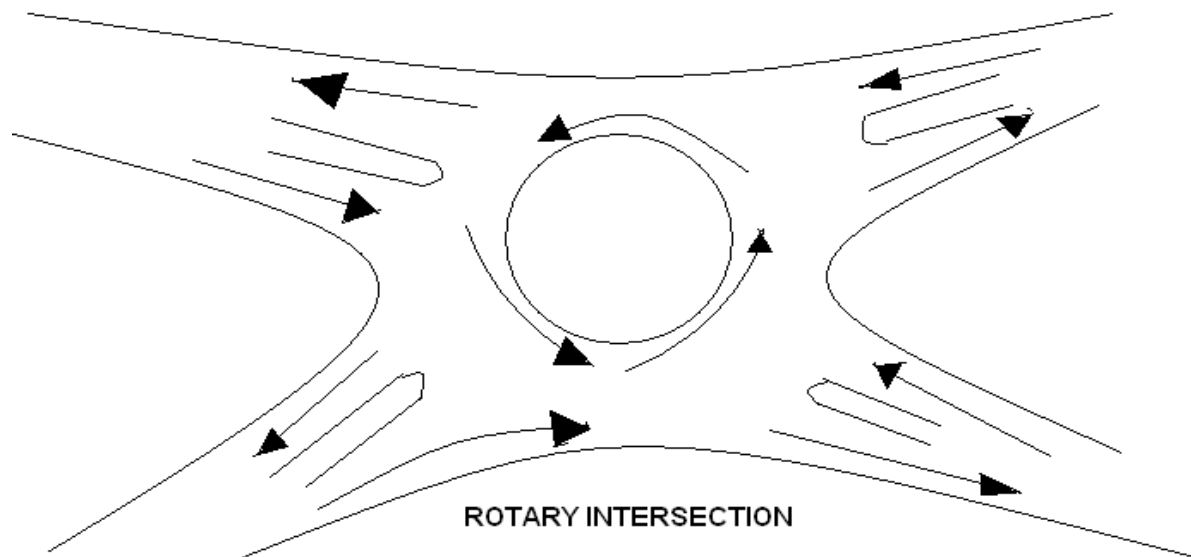
TEE (PARTIAL CHANNELIZATION)

The various advantages of channelized intersection are

- ✓ By channelization, vehicles can be confined to definite paths.
- ✓ Angle of merging streams can be forced to be flat angles so as to cause minimum disruption.
- ✓ Both major and minor conflict areas within the intersection can be considerable decreased.
- ✓ Angle between intersecting streams of traffic may be kept as desired in a favourable way.
- ✓ Speed control can be established over vehicle entering the intersection.
- ✓ Refuse islands can be provided for pedestrians within the intersection area.
- ✓ Points of conflict can be separated.
- ✓ The channelization island provides place for installation of signs and other traffic control devices.

### 3. ROTARY INTERSECTION OR ROUND-ABOUT

A rotary intersection or traffic rotary or round-about is an enlarged road intersection where all converging vehicles are forced to move round a large central island in one direction (anticlockwise direction) before they can weave out of the traffic flow into their respective directions radiating from the central island.



The main objectives of providing a rotary are to eliminate the necessity of stopping even for crossing streams of vehicles and to reduce the area of conflict. The crossing of vehicles is allowing all vehicles to merge into the stream around the rotary and then to diverse out to the radiating road. A rotary intersection may take one of the following shapes: circular, elliptical, turbine or tangent.

#### **ADVANTAGES OF A ROTARY INTERSECTION**

1. Crossing manoeuvre is converted into weaving or merging and diverging operations. Hence, there is no necessity of any vehicle, even those which have to go in cross sections, to stop and proceed within traffic rotary. Thus, the journey is more consistent and comfortable when compared within any other intersection at grade.
2. All traffic including those turning right or going straight across the rotary have equal opportunity as those turning left.
3. The variable cost of operation of automobile is less at a rotary than at a signalized intersection where the vehicles have to stop and proceed. Though the distance to be traversed by vehicles which are to turn to the left or proceed straight across is higher, still the fuel consumed in the process of crossing the rotary intersection is likely to be less.
4. Traffic police or signals are not necessary to control the traffic as the rotary could function by itself as traffic controlled intersection and the simplest of all controls. There is almost no maintenance cost.
5. The possible number of accidents and the severity of accidents are quite low because of low relatively speed. Further weaving, merging and diverging manoeuvres are easier and less

dangerous operations than crossing. Check on speed of vehicles is automatically enforced by proper design.

6. Rotaries can be constructed with advantage when the number of intersecting roads is between four and seven.
7. The capacity of the rotary intersection is the highest of all other intersection at grade. The rotary can accommodate a total traffic of about 3000veh/h and enable radial streets to carry traffic almost to their full capacity.

### **LIMITATIONS OF ROUND-ABOUT**

1. A round-about requires comparatively large area of land and so where space is limited and costly as in built up areas, the total cost may be very high.
2. Where pedestrian traffic is very high, as in urban areas the rotary by itself cannot control the traffic and hence has to be supplemented by traffic police. If the vehicular traffic has to stop to allow pedestrian to cross, then the main purpose of the rotary is defeated.
3. In places where there is mixed traffic and large number of cyclists and pedestrians, the design of rotary becomes too elaborated and operation and control of traffic also becomes complex.
4. Where the angle of intersection of two roads is too acute, or when there are more than seven intersecting roads, rotaries are unsuitable.
5. When the distance between intersections on an important highway is less, rotaries become troublesome.
6. When there is a large number of cycles and animals drawn vehicles, the extra length to be traversed by crossing and left turn traffic is considered troublesome and there is tendency to violate the traffic regulation of anti-clockwise movement around the central island.
7. When the traffic volume is very low, as in most rural areas, construction of rotary cannot be justified.

### **GRADE SEPERATED INTERSECTION OR INTERCHANGE**

An interchange is a grade separated in which vehicles moving in direction of flow may transfer direction by the use of connecting roadways called ramps.

This type of grade separated or intersection causes least delay and hazard to the crossing traffic and in general is much superior to intersections at grade from the view of traffic safety and efficient. It is achieved by means of vertical level. The grade separation may either be by over pass or under pass transform of route which is made possible by ramps. These ramps can be direct, semi-direct or indirect.

The direct ramp involves diverging to right side and merging from the right. Semi-direct interchange ramp allows diverging to the left but merging is from the right side. In the direct method of interchange ramp, a simple diverging to the left and a merging from the left side are involved which are simpler and less hazardous.

### **ADVANTAGES OF GRADE SEPERATED INTERSECTION**

1. Maximum facility is given to the crossing traffic. As the roads are separated this avoids necessity of stopping and avoids accidents while crossing.
2. There is increased safety for turning traffic and by direct interchange ramp even left turn is made quite easy and safe by converting into diverging to left and merging from left.
3. There is overall increase in comfort and convenience to the motorists and saving in travel time and vehicle operation cost.
4. The capacity of the grade operated intersection can practically approach that of the two cross roads.
5. Grade separation is essential part of controlled access highway like expressway and freeway.
6. It is possible to adopt grade separation for all likely angles and layout of intersecting roads.
7. Stage construction of additional ramps is possible after the grade separation structures between main roads are constructed.

### **DISADVANTAGES**

1. It is very costly to provide complete grade separation and interchanges facilities.
2. Where there is limited right of way like built up or urban area or where the topography is not favourable, construction of grade separation is costly, difficult and undesirable.
3. In flat or plain terrain, grade separation may introduce undesirable crests and sags in the vertical alignment.

### **INTERCHANGES**

Grade separation intersection with complete interchange facilities is essential to develop highway with full control access. When there is intolerable congestion and accidents at the intersection of two highways carrying very heavy traffic there is no better solution than to provide grade separation intersection.